ChE 382: Unit Operations Laboratory
GAS ABSORPTION
IN A PACKED BED
REVISION: Spring 2024v1

Introduction

Packed beds are widely used by chemical engineers to contact gas and liquid phases. Examples are gas absorption and distillation. They are also often used in chemical reactors where the catalyst is impregnated on the surface of the packing. For rational process design, chemical engineers need to be able to characterize the behavior of these contactors. Typically, one wants to be able to describe the pressure drop through the unit to allow sizing of the pumps. Additionally, it may be important to know the heat transfer between the bed and the wall if a highly exothermic or endothermic reaction is occurring. There are numerous packings used for these contactors and numerous calculations to allow estimation of operating parameters such as pressure drop. However, in many cases the correlations do not cover the desired range of operations and it is necessary to generate experimental data.

Packed Bed Flooding

An important decision for the engineer is the selection of relative gas and liquid flowrates in the packed-bed reactor. If the gas flow is much higher than the liquid flow, the gas tends to lift out of the bed and there is poor contact between the phases. Correlations have been developed to guide in process design. An important concept for these design tools is bed flooding. If the pressure drop is measured as a function of gas flowrate for a fixed liquid flowrate, a sharp break in slope will occur. This is defined as the point of flooding and is where the gas flowrate is sufficient to begin the holdup the liquid in the bed. Experience has shown that operating at 50% of flooding gives optimal gas-liquid contacting.

Objective

The objective of this experiment is:

1. To determine fluid flow/pressure drop characteristics in a packed bed gas absorber, including the flooding point.
2. To determine gas absorption characteristics and the overall mass transfer coefficient in the removal of CO_2 from an air stream with water in a packed bed gas absorber.

Theory

This is a basic mass-transfer process where the CO_2 is transported across the gas-liquid interface, so one can write a simple expression:

$$N = ka\Delta C \qquad (1)$$

where N is the moles of CO_2 transferred, ka is the mass transfer coefficient times the interfacial area (which we do not know) and ΔC is the driving force (partial pressure, concentration, mole fraction). There are several good references on packed bed and gas absorption principles and theory: Wankat [1], King [2], McCabe, Smith and Harriott [3], and Perry's Chemical Engineering Handbook [4] and Treybal [5].

Experiments

An Armfield Gas Absorption Column (UOP7) packed with 1-cm Raschig rings will be used for the experiments. A schematic diagram of the gas absorption column is shown in Figure 1. There is a liquid flow system using a centrifugal pump and a rotameter to determine the liquid flowrate. The system consists of an air supply from the house air and a supply of pure carbon dioxide from a high-pressure cylinder. Both gas streams have rotameters to determine the volumetric flowrate. The absorber system is constructed such that the air/carbon dioxide gas flows countercurrent to the liquid.

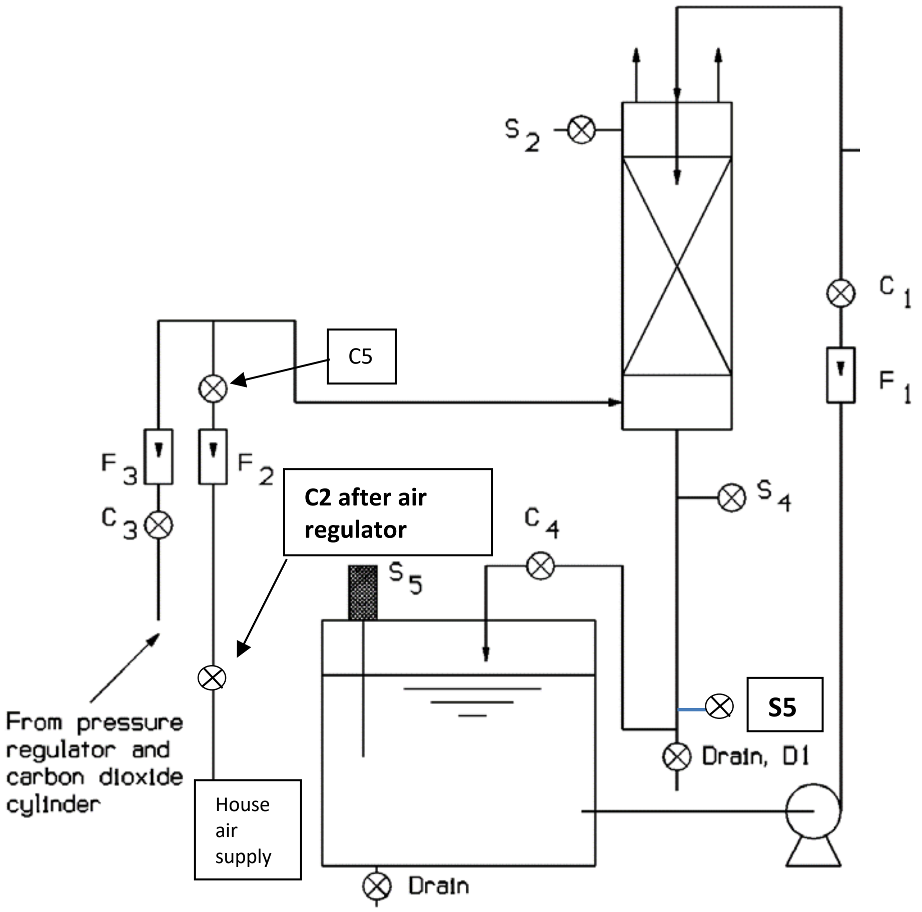

Figure 1. Gas Absorption Column System- Note: Valve C2 is in back of unit after pressure regulator.

The system has two, red oil manometers to determine the pressure drop across the absorber unit (use one for the pressure drop measurements).

The CO_2 composition in the air stream is determined volumetrically using a Hempl Apparatus located on the gas absorption panel. This analytical approach is based on determining the volume of CO_2 absorbed from a gas sample volume, and thus the carbon dioxide composition of the gas sample. Details of the Hempl Apparatus are provided in Appendix A.

The liquid-phase concentration of CO_2 is determined by titration. This is a classical analytical technique for determining concentrations of acids and bases in aqueous solution and is described in Appendix B.

Experimental Procedure

The experimental procedure is split into three separate parts.

Part 1: You will characterize the packed bed column and determine flooding for air/water flowrates**. It is recommended that you perform Part 1 in your Pre-Lab Lab Period.**

Part 2: You will study the absorption of CO_2 by water by first bringing the water into equilibrium with an airstream containing CO_2 in a closed-loop water system. After equilibrium is reached, you will study the desorption of CO_2 back into the airstream.

Part 3: You will study the absorption of CO_2 by water in a continuous, open-loop water system.

Part 1. Characterization of Liquid and Gas Flow in Packed Columns

Void Volume: Initially, the void volume of the packed bed should be determined. One method measures the amount of water necessary to immerse a given volume of packing. The void volume can then be obtained from the relationship:

$$\epsilon = \frac{V_{void}}{V_{Total}} \qquad (1)$$

where ε is the void volume defined as volume of voids / total volume of the packed bed material. Note that equation (1) is sensitive to the value of the porosity. The void volume of the solid phase includes the voids between the pieces of packing as well as the internal open spaces of each individual piece. It is these spaces through which the gas and liquid must pass. In practical applications, the packed bed material may have internal voids due to inherent porosity. This internal porosity is usually present in very fine pores (< 1 μm in diameter) and depending on the nature of the solid may represent either a negligible or significant fraction of the overall volume of the solid. For metals, glass, or a glazed ceramic, this internal porosity is normally absent. When the material has no appreciable fine pore volume, one can use water to measure the larger void spaces of the packed bed material or the void spaces of a bed of solids. For this experiment, this may be adequately done using a graduated cylinder filled with the packed bed material to a specific volume. Water is added until the voids are filled. Record the volume of water added.

Pressure Drop: In the second part of Part 1, the pressure drop across the packed bed is measured as a function of the air and water flow rates. The pressure drop across the packed bed is measured with the manometer connected to the column using the top and bottom pressure taps.

1. Close the drain valve under the liquid reservoir tank and the column drain valve, **D1**. Fill the liquid reservoir tank three-quarters full with deionized water (use blue hose from sink). Record the temperature of the water.
2. Open the compressed air shut-off valve **C2** in the back of the unit and adjust the air flow rate through flowmeter **F2** to 20 L/min using control valve **C5**. Make sure that **C3 is closed.** Allow air to flow through the column for 5 minutes to dry the packing. Then, measure the pressure drop across the bed for different gas flowrates with no liquid flow.
3. Close the gas flow control valve **C5** and start the liquid pump. Open valve **C4.** Open valve **C1** to expel air from the pump, then adjust the water flow to 1.5 L/min on flowmeter **F1** by adjusting flow control valve **C1**. Adjust valve **C4** so that the water level below the column is above the sample valve **S4** but below the air inlet tube. Suggestion: Opening sample valve **S4** and placing a bucket under the sample tubing to collect the water can help maintain the correct water level.
4. Open gas flow control valve **C5** and adjust flowmeter **F2** to 20 L/min. Measure the gas pressure drop as you increase the air flowrate until flooding occurs. Repeat the experiment for a several more water flowrates (3 L/min, 6 L/min, 8 L/min). **Note: Be careful when you approach the flooding point as the pressure drop becomes large. Watch manometer carefully and reduce the air flow at this point to prevent the red oil from being expelled out of the manometer into the tubing!!**

Part 2a: Absorption of CO_2 by H_2O in a Closed-Loop Water System

1. Close the drain valve under the liquid reservoir tank and the column drain valve, **D1**. Fill the liquid reservoir tank three-quarters full with deionized water using the blue hose from the sink. Record the temperature of the water. Collect a water sample and determine its initial CO_2 concentration by titration following the Analysis Procedure (Appendix B). Place the cover on the tank.
2. Open control valve **C4**. With gas flow control valves **C5** and **C3** closed, start the liquid pump, open valve **C1** a couple turns for 15-30 s to expel air from the pump, then adjust the water flow through the column to 6 L/min on flowmeter **F1** by adjusting flow control valve **C1**. Adjust control valve **C4** so that the water level below the column is above the sample valve **S4** and below the air inlet tube. Suggestion: Opening sample valve **S4** and placing a bucket under the sample tubing to collect the water can help maintain the correct water level.
3. Open the compressed air shut-off valve in the back of the unit and adjust the air flow rate through flowmeter **F2** to 20 L/min using control valve **C5**. Check to make sure that control valve **C3 is closed. NOTE:** Record the air pressure. Monitor and readjust the air flow rate during the experiment as it tends to drift. Note any readjustments on your data sheets.
4. Carefully open the main valve on the carbon dioxide cylinder all the way. Slowly adjust the pressure regulating valve on the carbon dioxide regulator to give a gauge reading of approximately 20 psi. Open the valve located just after the pressure regulator and heater block. Adjust valve **C3** to give a CO_2 flow rate on the flowmeter **F3** corresponding to the flow rate you have been **assigned by your faculty instructor, 2 through 5 L/min. Turn on the heater block switch.**
5. After 5 min of operation, take 50 mL water samples at appropriate intervals from **S4** until the dissolved CO_2 values are constant. Analyze the samples by titration following the directions in the Analysis Procedure (Appendix B).
6. **Measure the volume fraction of CO_2 in the air leaving the column (from sample port S2) once the dissolved CO_2 levels are constant using the Hempl apparatus (see Appendix A).**

Part 2b: Desorption of CO_2 from H_2O into air

1. Once you are done with Step 6 above, shut off the CO_2 flow at the flowmeter (valve **C3**), decrease the liquid flow rate to 1.5 L/min, open the drain valve, **D1**, under the column, and close the control valve, **C4**.
2. Adjust the drain valve, **D1**, under the column to maintain a liquid level above sample tap, **S4** but below the air inlet tube.
3. Immediately take water samples as quickly (e.g. 1 minute intervals) as possible from **S4**. Analyze the samples by titration following the directions in the Analysis Procedure (Appendix B).

Part 3: Absorption of CO_2 by H_2O in an open-loop water system

1. Close the drain valve under the liquid reservoir tank and the column drain valve, **D1**. Fill the liquid reservoir tank three-quarters full with deionized water using the blue hose from the sink. Record the temperature of the water. Collect a water sample and determine its initial CO_2 concentration by titration following the Analysis Procedure (Appendix B). Place the cover on the tank.
2. Open control valve **C4**. With gas flow control valves **C5** and **C3** closed, start the liquid pump, open valve **C1** a couple turns for 15-30 s to expel air from the pump, then adjust the water flow through the column to 1.5 L/min.
3. Open the column drain valve, **D1**. and close control valve **C4**. Use the column drain valve, **D1**, to maintain a liquid water height below the air/CO2 inlet but above the sampling valve **S4**. It is much easier to maintain a liquid level if sample valve **S4** is left open. A bucket can be placed under the sample tubing to collect the water.
4. Open valve **C5** and adjust the air flow rate through flowmeter **F2** to a value of 20 L/min. **NOTE:** Monitor and readjust the air flow rate during the experiment as it tends to drift.
5. Carefully open the main valve on the carbon dioxide cylinder all the way. Slowly adjust the

pressure regulating valve on the carbon dioxide regulator to give a gauge reading of approximately 20 psi. Open the valve located just after the pressure regulator and heater block. Adjust valve **C3** to give a CO_2 flow rate on the flowmeter **F3** corresponding to the flow rate you have been assigned by your faculty instructor, 2 through 6 L/min. **Turn on the heater block switch.**

6. As soon as the CO_2 is turned on, take 50 mL water samples from **S4** at 30 second intervals during the first two minutes. Increase the interval between samples to one minute for the next five minutes. The interval can then be increased to five minutes until the concentration reaches a steady state. Analyze the samples following the directions in the Analysis procedure (Appendix B).
7. **Analyze a gas sample from sample port S2 to measure the volume fraction of CO_2 in the outlet air once steady state is reached using the Hempl Apparatus (see Appendix A).**

Note: Steps 2, 3, and 4 should be done as quickly as possible or the flow rates pre-set, the reservoir again filled with water and the experiment started. This will ensure that you have enough water to complete the experiment.

Data Analysis

The Results and Discussion sections should include:

1. Packed Bed Characterization: Make a plot of pressure drop versus water and air flowrate. Show the flooding point on the graph. Show the porosity obtained from your measurement. Describe the various regimes of gas-liquid absorbers and how the pressure drop versus gas flowrate can be used to quantitatively define the various regimes. The discussion should also include the region selected for optimal gas absorber operation and the reasons for this selection.
2. Absorption/Desorption of CO_2 into Water: Make plots of your CO_2 concentrations versus time from the Part 2a, 2b and Part 3 experiments. Describe the CO_2 concentration initially in the air and at the outlet of the column (Hempl apparatus measurements). Make a table of the mass-transfer coefficients that you obtained. Since this is a basic mass-transfer process where the CO_2 is transported across the gas-liquid interface, explain how the data are used to describe the absorption and desorption process in terms of mass-transfer coefficients. There is a mass-transfer coefficient for the transport from the gas phase to the gas-liquid interface and another one for the transport from the interface to the bulk of the liquid. See references for a discussion of these resistances in series. You can show that one of the two resistances dominates the transport process.

SAFETY NOTES

Before starting the experiment, review the Material Safety Data Sheets (MSDS) on NaOH, phenolphthalein, compressed air, and carbon dioxide. The sheets can be found in the MSDS notebook located in the laboratory.
Disposable nitrile gloves should be worn when handling the NaOH solutions.
Check the safety requirements you need to be aware of when using high pressure gas cylinders.
A small amount of NaOH spilled on the outside of glassware can cause the glassware to be extremely slippery when wet. Be careful.

WASTE DISPOSAL PROCEDURES

Titration waste from this experiment should be placed in the Titration Waste bottles located in the South-wall fume hood.

References

[1] Wankat, P.C., Chapter 13, Chapter 15, “Equilibrium Staged Separations”, Prentice-Hall, Inc. 1988.

[2] King, C.J., Chapter 12, “Separation Processes”, McGraw-Hill, 1980.

[3] McCabe, W. L., Smith, J. C., Marriott, P., Chapter 18, "Unit Operations of Chemical Engineering", 7th Edition, McGraw-Hill, 2005.

[4] R. H. Perry and D. W. Green, Eds., *Perry’s Chemical Engineer’s Handbook*, 7th ed. McGraw-Hill, 1997.

[5] R. Treybal, *Mass Transfer Operations*, 2nd ed. McGraw-Hill, 1980.

Appendix A

Use of Gas Analysis Equipment

Measurement of the Volume Fraction of CO2 in the Gas Stream

Familiarize yourself with the analysis system for CO2 in the gas stream; this is described below. The TA or Faculty Instructor will show you how it works. (The absorption globes should already be filled with 1.0 N NaOH.) It is also important for you to understand the operation of the three-way stopcocks; diagrams are given below.

Steps to become familiar with the Hempl Apparatus:

1. Safety: What are the safety precautions that must be taken when working with high pressure gas cylinders?

2. Try out the analysis system by measuring the volume fraction of CO_2 in the ambient air. What do you expect the volume fraction to be? Are the results reasonable?

3. Insure the compressed air line is connected to the unit and the valve on the air line is open (next to the unit). Adjust the air flow rate through the flow meter to 20 L/min using the control valve **C2**. The air flow tends to drift so it is important to monitor it periodically.

4. Reconnect the CO_2 line to the flow meter at the rear of the experiment panel. Adjust the CO_2 flow to 2 L/min.

5. Collect a sample of gas from the top of the column, valve **S2**, and determine the volume fraction of CO_2 in it. Repeat the measurement with a new gas sample.

6. You should continue to measure the volume fraction of CO_2 in gas samples from the top of the column for CO_2 flow rates of 3, 4 & 5 L/min with the air flow set at 20 L/min throughout.

In the following procedure three-way stopcocks are used in sampling the gas stream. The operation of the three-way stopcock is shown in Figure 2.

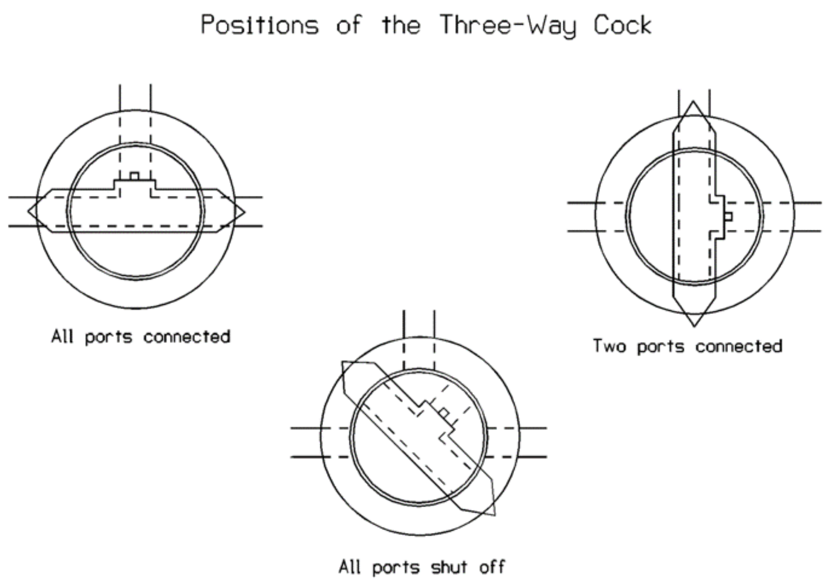

Figure 2. Three-way stopcock positions

Gas Sampling Procedure

1. Fill the two globes of the absorption analysis equipment on the left of the panel with 1.0 N NaOH solution if it has not already been done. Use a small funnel and tubing. NOTE: Wear safety gloves and goggles. Adjust the level in the globes to the "0" mark on the sight tube, by draining liquid through valve Cv into a flask (See A in Figure 3.). This will normally have been done for you. Ask the TA or Instructor to add more NaOH if necessary to bring the level to zero on the measuring tube scale.

2. After 15 minutes or so of steady operation, take samples of gas from sample point S2. Analyze these samples as given in the instructions listed below.

3. Pull the piston of the syringe out to the stop. Take a squirt bottle of deionized water and wet the piston of the syringe. This insures there is a water seal so the gas pulled into the syringe can not leak out to the atmosphere around the syringe piston. This may have to be done several times during the experiment since the water on the piston does evaporate.

4. Flush the sample lines by repeated sucking from the line using the syringe and expelling the contents of the syringe to the atmosphere. Note that the volume of the cylinder is about 100 mL. Estimate the volume of the tubing leading to the syringe. How do you estimate this volume and how many times will you need to purge the line (Steps B & C)?

5. With the absorption globe isolated and the vent to atmosphere closed, fill the syringe from the selected line by drawing the syringe piston out slowly (Step B). Note volume taken into the syringe V1, which should be approximately 40 mL for this particular experiment See WARNING note below). Wait at least two min to allow the gas to come to the temperature of the syringe.
6. Isolate the syringe from the column and the absorption globe and vent the syringe to atmospheric pressure. Close after about 10 seconds. (Step D).

7. Connect the syringe to the absorption globe. The liquid level should not change. If it does change, briefly open to atmosphere again.

8. Wait until the level in the indicator tube is on zero showing that the pressure in the cylinder is atmospheric.

9. Slowly close the piston to empty the gas in the syringe into the absorption globe. Slowly draw the piston out again (Steps E and F). Why would you want to do this step slowly?

Note the level in the indicator tube.

Repeat steps E and F until no significant change in level occurs. Read the indicator tube marking - V2. This represents the volume of the gas sampled.

WARNING: If the concentration of CO_2 in the gas sampled is greater than 8%, it is possible to suck liquid into the cylinder. This will ruin your experiment and takes time to correct. Under these circumstances, do not pull the piston out to the end of its travel. Stop it at a particular mark, e.g. V1 = 40 on the coarse scale, and read the fine scale.

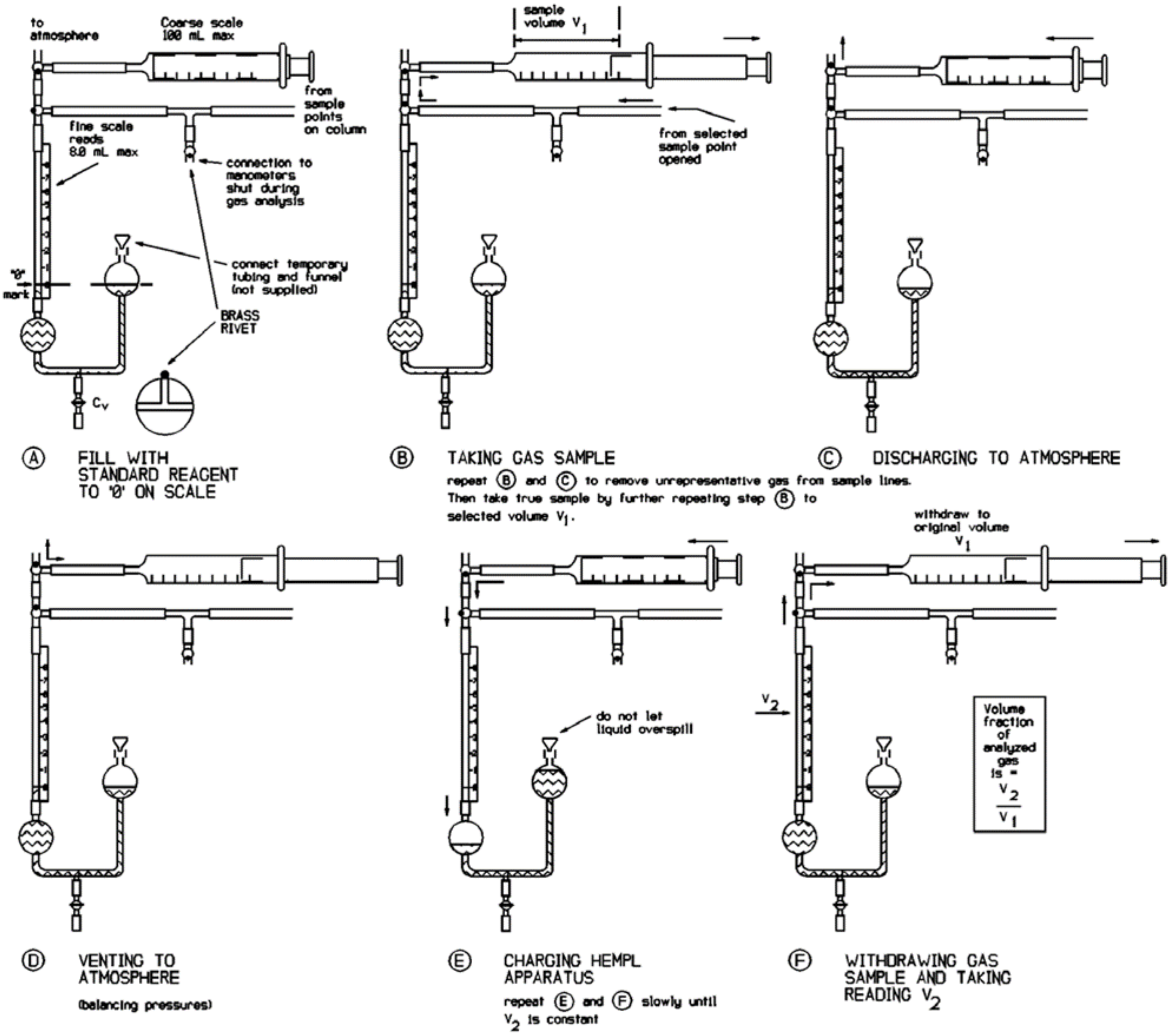

Figure 3. Hempl apparatus for gas analysis

Appendix B

ANALYSIS PROCEDURE FOR CO2 IN WATER

1. The sodium hydroxide used in the experiment is standard solution of either 0.01 N or 0.05 N NaOH.

2. Collect 50 mL with a graduated cylinder from the liquid outflow point S4. If needed, collect a sample from the reservoir tank. NOTE: Purge the sample line before collecting sample. The sample should be collected with the sample tube below the liquid level in the graduated cylinder.

3. Place the sample in a 125 mL erlenmeyer flask.

4. Add 3-5 drops of phenolphthalein indicator solution. If the sample turns red immediately, no free CO2 is present. If the sample remains colorless, swirl the flask and titrate with standard NaOH solution. The end point is reached when a definite pink color persists for about 30 seconds - record volume VB of the NaOH added.

5. The amount of free CO_2 in the water sample is calculated from:

$$C_d\left(\frac{gram\ mole}{liter\ of\ free\ CO2}\right) = \left(\frac{V_B \times (Conc.Base)}{mL\ of\ Sample}\right)$$

C_{di} = Concentration at inlet (corresponds to conditions at the top of the tower, S6).

C_{do} = Concentration at outlet (corresponds to conditions at the bottom of the tower, S4).

Note: Solubility of CO_2 in water is a strong function of temperature and pH.